SOLUTIONS

Question 1

(a)
Quantum confinement in the spherical nanocrystal will raise the bottom of the conduction band and lower the top of the valance band by the corresponding confinement energy $E_k = [\pi^2\hbar^2]/[2m_{eff}m_0(d/2)^2]$, where $\hbar = h/2\pi$ is reduced Planck's constant, d is the nanocrystal diameter, m_0 is the electron rest mass, and m_{eff} is the relevant effective mass. This increases the band gap E_g. Confinement of electrons within a small volume, from Heisenberg's Uncertainty Principle, will remove the requirement for momentum conservation in the *e-h* recombination process.

The energy E_p of the emitted photon is then given by:

$$E_p = \hbar\omega_p = E_g + [\pi^2\hbar^2]/[2m_em_0(d/2)^2] + [\pi^2\hbar^2]/[2m_hm_0(d/2)^2]$$
$$= 1.1 + 0.375 + 0.66 = 2.135 \text{ eV} = 3.416 \times 10^{-19} \text{ J}$$

[3]

The wavelength of the emitted photon is then given by:

$$\lambda = c/f = 2\pi c/\omega_p = 2\pi c\hbar/E_p = 582 \text{ nm}.$$

[2]

If the answer uses the confinement energy $E = [\pi^2\hbar^2]/[2m_{eff}m_0d^2]$ for 1-D, then 1 mark may be deducted.

(b)
From Figure 1.1, $H = t_1 + t_2 = 5$ nm + 10 nm = 15 nm. The gate capacitance is then given by:

$$C_g = \frac{2\pi\varepsilon_r\varepsilon_0 L}{ln\frac{2H}{R}} = \frac{2\pi \times 3.9 \times 8.854 \times 10^{-12} \times 0.5 \times 10^{-7}}{ln\frac{2 \times 15}{10}} = 0.98 \times 10^{-17}\ F$$

At threshold voltage $V_{th} = 0.4$ V, the charge induced in the nanowire is given by:

$$Q = C_gV_{th} = 0.98 \times 10^{-17} \times 0.4 = 0.39 \times 10^{-17} \text{ C}$$

We can then find the total number of charge carriers in the nanowire:

$$N = Q/e = 0.39 \times 10^{-17} / 1.6 \times 10^{-19} = 24 \text{ carriers}.$$

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Therefore, the carrier concentration is:

$$n = N/\text{volume} = N/(\pi R^2 \times L) = 24/(\pi \times 10 \times 10 \times 10^{-18} \times 10^{-7})$$
$$= 7.6 \times 10^{23}\ \text{m}^{-3}$$

[5]

(c)
For angles of normal incidence 64.3° and 30°, the corresponding angles between the X-rays and the atomic planes are $\theta_1 = 90 - 64.3^o = 25.7^o$ and $\theta_2 = 90 - 30^o = 60^o$.

Using Bragg's Law, $n\lambda = 2d \sin\theta$, we then have for the two angle values:

$d = n_1\lambda/2 \sin\theta = n(0.3)/(2 \sin 25.7) = n_1(0.346)$

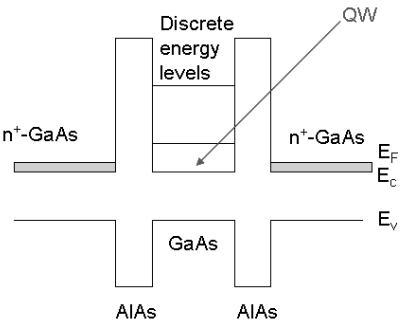
and,

$d = n_2\lambda/2 \sin\theta = n(0.3)/(2 \sin 60) = n_2(0.173)$

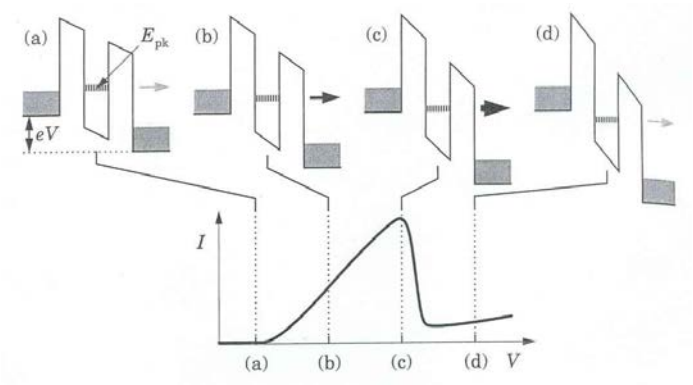
These two equations give the same value of $d = 0.346$ nm for $n_1 = 1$ and $n_2 = 2$. We then have an atomic plane separation $d = 0.346$ nm.

[5]

(d)
Answers should consider the points below:
Resonant tunnelling diodes typically use double-barrier heterostructures to form a potential well along one dimension. For example, in a n^+GaAs/AlAs/i-GaAs/AlAs/n^+GaAs heterostructure, the AlAs layers form tunnel barriers and the central GaAs region forms a potential well. A simplified band diagram is shown below:



The discrete states in the well lead to current peaks in the *I-V* characteristics, as they drop below the conduction band edge in the source. Current flows when a state overlaps with filled states in the source. The process is shown below diagrammatically, for a single current peak/state in the well:



[5]

Question 2

Answer:
(a)

$$\psi_{100} = A\exp\left(-\frac{Zr}{a_0}\right)$$

Normalising in spherical coordinates:

$$\int_{r=0}^{\infty}\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi}\psi_{100}^2\, r^2 \sin\theta\, dr\, d\theta\, d\phi = 1 \qquad [3]$$

$$\Rightarrow A^2\int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} e^{-\frac{2Zr}{a_0}} r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$= A^2\int_0^{\infty} e^{-\frac{2Zr}{a_0}} r^2 dr \int_0^{\pi}\sin\theta\, d\theta \int_0^{2\pi} d\phi \qquad [2]$$

$$= A^2 \left|-\cos\theta\right|_0^{\pi} \cdot \left|\phi\right|_0^{2\pi} \cdot \left(\left| \frac{r^2 e^{-2Zr/a_0}}{-\frac{2Z}{a_0}} \right|_0^{\infty} - \int_0^{\infty} 2r \frac{e^{-\frac{2Zr}{a_0}}}{-\frac{2Z}{a_0}} dr \right)$$

$$= -A^2(-1-1)\cdot(2\pi - 0)\cdot\left(+\frac{2a_0}{2Z}\int_0^{\infty} r e^{-\frac{2Zr}{a_0}} dr\right)$$

$$= +8\pi A^2\left(\frac{a_0}{2Z}\left(\left| \frac{r e^{-2Zr/a_0}}{-\frac{2Z}{a_0}} \right|_0^{\infty} - \int_0^{\infty} \frac{e^{-2Zr/a_0}}{-2Z/a_0} dr\right)\right)$$

$$= +8\pi A^2\left(\frac{a_0}{2Z}\left(+\frac{a_0}{2Z}\left| \frac{e^{-\frac{2Zr}{a_0}}}{-\frac{2Z}{a_0}} \right|_0^{\infty}\right)\right)$$

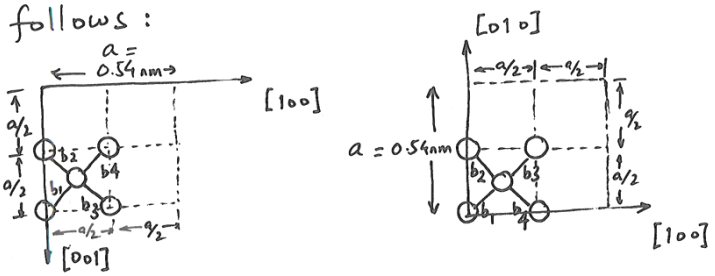
$$= +8\pi A^2\left(\frac{a_0}{2Z}\right)^2 \cdot \left(-\frac{a_0}{2Z}(0-1)\right)$$

$$= +8\pi A^2\left(\frac{a_0}{2Z}\right)^3 = 1$$

$$\Rightarrow \quad A = \sqrt{\frac{1}{\pi}}\left(\frac{Z}{a_0}\right)^{3/2} \qquad [5]$$

Question 2(b)

Looking from the top, i.e [010] direction, and from the front, i.e. [001] direction, the atoms are arranged as follows:



(i) Bond Lengths are the same and are given by:

$$|b| = \sqrt{\left(\frac{a}{4}\right)^2 + \left(\frac{a}{4}\right)^2 + \left(\frac{a}{4}\right)^2} = \sqrt{3\left(\frac{a}{4}\right)^2}$$

$$= \frac{\sqrt{3}}{4} a = 0.43 \times 0.54 = 0.23 \text{ nm}$$

[4]

(ii) The fundamental direction vectors, parallel to the bonds, can be written down by inspection:

Bond 'b_1': $[\bar{1}\,\bar{1}\,1]$

Bond 'b_2': $[\bar{1}\,1\,\bar{1}]$

Bond 'b_3': $[1\,1\,1]$

Bond 'b_4': $[1\,\bar{1}\,\bar{1}]$

[4]

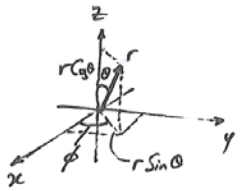
Question 2(c)

$$|2S\rangle = \psi_{200} \sim \left(1 - \frac{Zr}{2a_0}\right) e^{-Zr/2a_0}$$

$$|2P_0\rangle = \psi_{210} \sim e^{-Zr/2a_0} . r\cos\theta$$

$$|2P_{\pm1}\rangle = \psi_{21,\pm1} \sim e^{-Zr/2a_0} . e^{\pm i\phi} r\sin\theta$$

The $|2p\rangle$ states can be rewritten as (see diagram):



$$|2P_z\rangle = \psi_{210} \sim r\cos\theta\, e^{-Zr/2a_0}$$

$$= z\, e^{-Zr/2a_0}$$

↖ axis

[2]

and

$$|2P_{\pm1}\rangle = \psi_{21,\pm1} = e^{-Zr/2a_0} e^{\pm i\phi} r\sin\theta$$

$$= e^{-Zr/2a_0} (r\sin\theta)(\cos\phi \pm i\sin\phi)$$

$$= e^{-Zr/2a_0} (x \pm iy)$$

We may then construct the linear combinations:

$$|2P_x\rangle = \frac{\psi_{21,1} + \psi_{21,-1}}{2} = x e^{-Zr/2a_0}$$ [2]

$$|2P_y\rangle = \frac{\psi_{21,1} - \psi_{21,-1}}{2i} = y e^{-Zr/2a_0}$$ [2]

The bond 'b_3' points along the [111] fundamental direction. We may construct further linear combinations, using the $|2s\rangle$, $|2p_x\rangle$, $|2p_y\rangle$ and $|2p_z\rangle$ states:

First combination:

$$|x+y+z\rangle = |2p_x\rangle + |2p_y\rangle + |2p_z\rangle$$
$$= (x+y+z)\,e^{-Zr/2a_0}$$

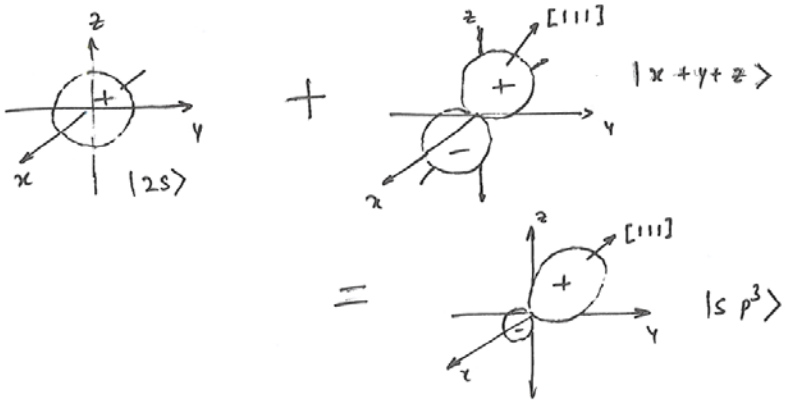
This is aligned along the [111] direction

We then construct the combination:

$$|sp^3\rangle = |2s\rangle + |x+y+z\rangle$$
$$= \left(1 - \frac{Zr}{2a_0}\right)e^{-Zr/2a_0} + (x+y+z)\,e^{-Zr/2a_0}$$ [3]

This points along the [111] direction

The wavefunction is plotted below:



Similarly, for the remaining bonds:

b_1 : $|2s\rangle + |-x-y+z\rangle$

b_2 : $|2s\rangle + |-x+y-z\rangle$

b_4 : $|2s\rangle + |x-y-z\rangle$ [3]

In the above, the answer should show the derivation of an expression for at least one of the bonds b_1-b_4 in detail, the rest may be written directly.

Question 3:

Answer:

(a) The electron wave vectors are as follows:

$$k_1 = \sqrt{\frac{2m_1 E}{\hbar^2}}$$

[2]

$$k_2 = \sqrt{\frac{2m_2 (V_0 - E)}{\hbar^2}}$$

[2]

(b) The boundary transmission matrices Γ_1 and Γ_2 are related to the electron wave function amplitudes as follows:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{2k_2} \begin{pmatrix} k \exp\left(k^* \frac{a}{2}\right) & k^* \exp\left(k \frac{a}{2}\right) \\ k^* \exp\left(-k \frac{a}{2}\right) & k \exp\left(-k^* \frac{a}{2}\right) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \Gamma_1 \begin{pmatrix} A \\ B \end{pmatrix}$$

[2]

$$\begin{pmatrix} C \\ D \end{pmatrix} = \frac{i}{2k_1} \begin{pmatrix} -k \exp\left(k^* \frac{a}{2}\right) & k^* \exp\left(-k \frac{a}{2}\right) \\ k^* \exp\left(k \frac{a}{2}\right) & -k \exp\left(-k^* \frac{a}{2}\right) \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \Gamma_2 \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

[2]

(c) To find Γ_{single}, we first note that:

$$\begin{pmatrix} C \\ D \end{pmatrix} = \Gamma_2 \Gamma_1 \begin{pmatrix} A \\ B \end{pmatrix} = \Gamma_{single} \begin{pmatrix} A \\ B \end{pmatrix} \Rightarrow$$

$$\Gamma_{single} = \frac{i}{2k_1} \frac{1}{2k_2} \begin{pmatrix} -k \exp\left(k^* \frac{a}{2}\right) & k^* \exp\left(-k \frac{a}{2}\right) \\ k^* \exp\left(k \frac{a}{2}\right) & -k \exp\left(-k^* \frac{a}{2}\right) \end{pmatrix} \begin{pmatrix} k \exp\left(k^* \frac{a}{2}\right) & k^* \exp\left(k \frac{a}{2}\right) \\ k^* \exp\left(-k \frac{a}{2}\right) & k \exp\left(-k^* \frac{a}{2}\right) \end{pmatrix}$$

[4]

This may be simplified as follows:

$$\Gamma_{single} = \frac{i}{2k_1}\cdot\frac{1}{2k_2}\begin{pmatrix} -k\exp\left(k^*\frac{a}{2}\right) & k^*\exp\left(-k\frac{a}{2}\right) \\ k^*\exp\left(k\frac{a}{2}\right) & -k\exp\left(-k^*\frac{a}{2}\right) \end{pmatrix}\begin{pmatrix} k\exp\left(k^*\frac{a}{2}\right) & k^*\exp\left(k\frac{a}{2}\right) \\ k^*\exp\left(-k\frac{a}{2}\right) & k\exp\left(-k^*\frac{a}{2}\right) \end{pmatrix}$$

$$= \frac{i}{4k_1k_2}\begin{pmatrix} -k^2\exp\left(k^*a\right)+\left(k^*\right)^2\exp\left(-ka\right) & -|k|^2\exp\left(k_2a\right)+|k|^2\exp\left(-k_2a\right) \\ |k|^2\exp\left(k_2a\right)-|k|^2\exp\left(-k_2a\right) & \left(k^*\right)^2\exp\left(ka\right)-\left(k\right)^2\exp\left(-k^*a\right) \end{pmatrix}$$

Here, we have used $kk^* = |k|^2$ (2 marks for seeing this). Remaining 4 marks to be awarded for the derivation of the matrix elements. The magnitude terms should be indicated clearly.

[6]

(d) The determinant for Γ_{single} is calculated as follows:

$$\det\left[\Gamma_{single}\right] = \left(\frac{i}{4k_1k_2}\right)^2\left(\left(-k^2\exp\left(k^*a\right)+\left(k^*\right)^2\exp\left(-ka\right)\right)\left(\left(k^*\right)^2\exp\left(ka\right)-\left(k\right)^2\exp\left(-k^*a\right)\right)\right.$$
$$\left.-\left(|k|^2\exp\left(k_2a\right)-|k|^2\exp\left(-k_2a\right)\right)\left(-|k|^2\exp\left(k_2a\right)+|k|^2\exp\left(-k_2a\right)\right)\right)$$
$$= \left(\frac{i}{4k_1k_2}\right)^2\left(-|k|^4\exp\left(2k_2a\right)+\left(k\right)^4+\left(k^*\right)^4-|k|^4\exp\left(-2k_2a\right)\right.$$
$$\left.+|k|^4\exp\left(2k_2a\right)-|k|^4-|k|^4+|k|^4\exp\left(-2k_2a\right)\right)$$
$$= \left(\frac{i}{4k_1k_2}\right)^2\left(\left(k\right)^4+\left(k^*\right)^4-2|k|^4\right)$$

Here, award 2 marks for the basic expression for the determinant (1st line). The multiplying factor for T_{single} must not be left out through the calculation (2 marks for seeing this). Remaining 6 marks to be awarded for the derivation of the expression above.

[10]

Hence, by using the expression given in the question, we find:

$$\det\left[\Gamma_{single}\right] = \left(\frac{i}{4k_1k_2}\right)^2\left(\left(k\right)^4+\left(k^*\right)^4-2|k|^4\right) = \left(\frac{-1}{16k_1^2k_2^2}\right)\left(-16k_1^2k_2^2\right) = 1$$

[2]

Answer, Question 1:

e) For a D = 525 μm-thick wafer, the base of the trapezoidal etch pit needs to have a side length $W_b = 1000 + \sqrt{2} \times D = \ 1742$ μm. The mask opening needs to be smaller than this by an amount $\delta = \frac{\sqrt{6}D}{S} = \sqrt{6} \times \frac{525}{35} = 36.7$ μm, so the required mask opening side length is $W_m = W_b - \delta = \mathbf{1705.7\ \mu m.}$

Reversing this calculation, the membrane side length W is given by:

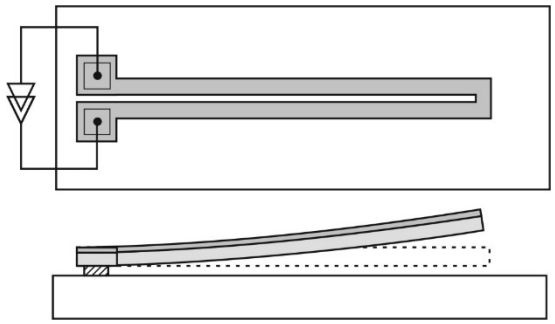
$$W = W_m + \frac{\sqrt{6}D}{S} - \sqrt{2}D$$

Differentiating we get: $$\Delta W \approx \left(\frac{\sqrt{6}}{S} - \sqrt{2}\right)\Delta D - \frac{\sqrt{6}D}{S^2}\Delta S$$

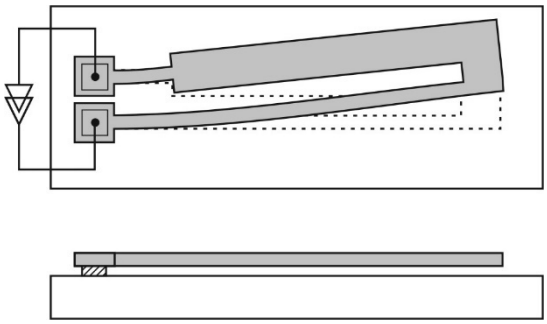
Both partial derivatives are <0, so the smallest membranes will correspond to D = 550 μm, ΔS = 40, and the largest membranes to D = 500 μm, S = 30. Substituting these values gives $\mathbf{962\ \mu m \ \leq W \leq 1039\ \mu m}$. [5]

f) Electrothermal actuators use differential thermal expansion between different parts of a structure to generate motion. In a material bimorph actuator, the differential thermal expansion occurs between two dissimilar materials that are in intimate contact, while in a shape bimorph actuator the device geometry leads to temperature differentials between different parts of a single mechanical layer. Heating is generally achieved by passing a current through either the mechanical layer or a deposited heater layer.

Material bimorph:



Shape bimorph:



Material bimorph actuators generally produce out-of-plane motion, while shape bimorph generally produce in-plane motion. Shape bimorph actuators are easier to fabricate because they can be realised in a single layer. They are also less prone to unwanted deflections due to intrinsic stress and ambient temperature variations. [5]

g) From the *Information for Candidates*, the Q is given by $Q = \frac{m\omega_0 d}{A\mu}$. Under scenario (i), the scaling laws for m, ω_0, d, A are L^3, L^{-1}, L^1, L^2, so the Q will scale as L^1. The scaling behaviours of ω_0 and A are the same under scenario (ii), while the scaling laws for m and d change to L^2 and L^0 respectively, so in this case Q will scale as L^{-1}. [5]

h) The arc length measured along the neutral axis of the deflected flexure is:

$$s = \int_0^l (1 + v'^2)^{1/2} dx \approx l + \frac{1}{2}\int_0^l v'^2 dx$$

The axial strain is therefore given approximately by:

$$\varepsilon \approx \frac{1}{2l}\int_0^l v'^2 dx$$

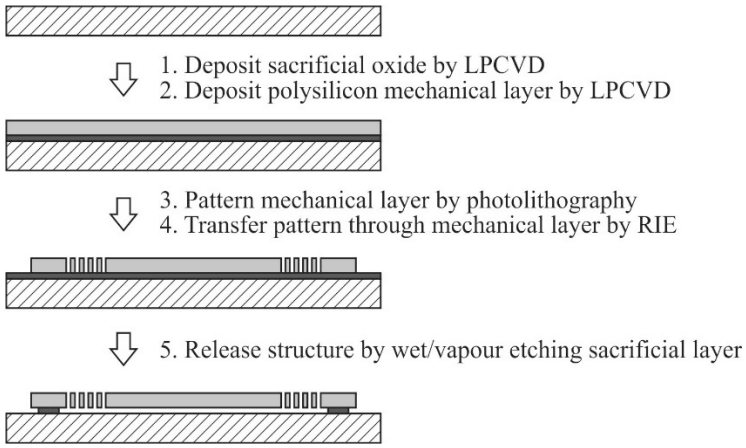
From the *Information for Candidates* we have $v(x) = \frac{Pl^3}{12EI}(3u^2 - 2u^3) = v_l(3u^2 - 2u^3)$ where $u = x/l$, so $v'(x) = \frac{6v_l}{l}(u - u^2)$, and:

$$\varepsilon \approx \frac{18{v_l}^2}{l^2}\int_0^1 (u^2 - 2u^3 + u^4) du = \frac{18{v_l}^2}{l^2}\frac{1}{30} = \frac{3}{5}\left(\frac{v_l}{l}\right)^2$$

[5]

Answer, Question 4:

a) This is a standard surface micromachining process with one mechanical layer:



[6]

b) If each serpentine spring is considered as four flexures in series then the stiffness of each spring will be one quarter of that of a single flexure. The overall stiffness of the suspension will then be that of a single flexure, i.e. $k = 12EI/L^3 = Ewh^3/L^3$. With E = 170 GPa, w = 20 μm, h = 5 μm, L = 400 μm we obtain k = 6.64 N/m. [6]

c) The capacitance between the proof mass and the substrate is $C = \frac{A\varepsilon_0}{g}$, and only g varies, so the sensitivity is $S = \frac{dC}{da} = \frac{dC}{dg}\frac{dg}{da}$. In steady state the gap is given by $k(g_0 - g) = ma$ where g_0 is the gap at zero acceleration. Combing these relations gives:

$$S = \left(-\frac{A\varepsilon_0}{g^2}\right)\left(-\frac{m}{k}\right) = \frac{mA\varepsilon_0}{kg^2}$$

Putting $A = 10^{-6}$ m^2, $m = \rho Ah = 1.165 \times 10^{-8}$ kg, k = 6.64 N/m, g = 2 μm gives S = 3.88 fF/ms^{-2} or 38 fF/g. [6]

d) The quick way to solve this problem is just to treat the downward shift of the force-displacement curve as a correction to the initial gap, g_0. Under an applied acceleration a, the effective initial gap becomes $(g_0 - \frac{ma}{k})$, and the formula for the snap-down voltage changes to:

$$V = \sqrt{\frac{8k(g_0 - \frac{ma}{k})^3}{27A\varepsilon_0}}.$$

Rearranging this to make a the subject gives:

$$a = \frac{k}{m}\left[g_0 - \sqrt[3]{\frac{27A\varepsilon_0 V^2}{8k}}\right].$$

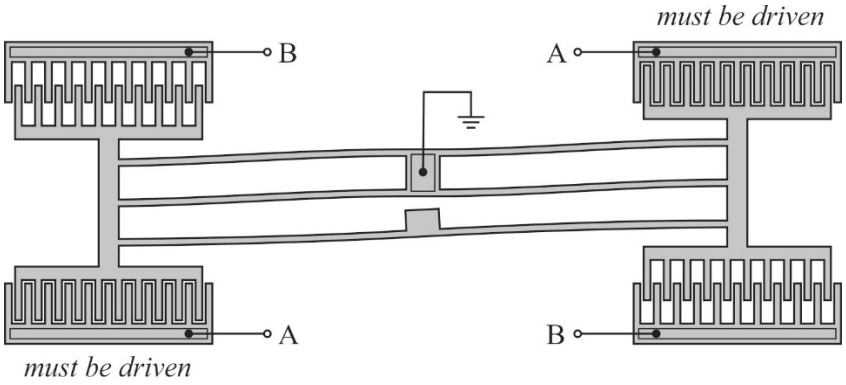
The acceleration at which the applied voltage will lead to snap down is obtained by putting V = 1 V, which gives a = 199 ms^{-2} or 20 g.

The problem can also be solved by straightforward application of snap-down analysis in the case where there is an additional static force due to the applied acceleration. [8]

e) The flexures in the serpentine springs will be subject to a torque which tends to twist them, increasing the overall end displacement for a given force. [4]

Answer, Question 5:

a) The upper four beams in the diagram form a pair of portal frames. These are pulled in opposite directions so as to deflect a longer flexure at the bottom of the diagram such that there is rotation but no translation at the centre, where the mirror is mounted. Electrostatic actuators can only pull in, so to produce the deflection shown the "A" actuators must be driven.



[6]

b) The stiffness seen by the actuators on each side is that of two flexures of length *l*, and a cantilever of length (*l* + *m*/2) in parallel:

$$k = 2 \times \frac{12EI}{l^3} + \frac{3EI}{(l+\frac{m}{2})^3} = \frac{w^3 hE}{4l^3}\left[8 + \frac{1}{\left(1+\frac{m}{2l}\right)^3}\right]$$

With w = 5 μm, h = 50 μm, l = 500 μm, m = 50 μm this gives *k* = 18.8 N/m. [8]

c) The capacitance in each electrode gap is $lh\varepsilon_0/g$ where *l* is the overlap length, and displacement is in the *l*-direction, so the force due to each gap has magnitude $\frac{\varepsilon_0 h V^2}{2g}$. Considering the LHS, and taking the +y-direction as up the page, the total force when both comb drives are being driven will be:

$$F_y = +\frac{N\varepsilon_0 h}{2g}(V_B^2 - V_A^2) = +\frac{N\varepsilon_0 h}{g}\left(\frac{V_B+V_A}{2}\right)(V_B - V_A).$$

By inspection the force on the other side will be equal and opposite. [6]

d) The bending equation is $M = EIv'' = P(L - x)$. Integrating once gives $v' = \frac{P}{EI}(Lx - \frac{x^2}{2})$, and integrating a second time gives $v = \frac{P}{EI}(\frac{Lx^2}{2} - \frac{x^3}{6})$. From these relations we can see that the end deflection and end rotation are given by $v_L = \frac{PL^3}{3EI}$ and $\theta = \frac{PL^2}{2EI} = \frac{3v_L}{2L}$. [6]

e) With $V_0 = V_D$ = 100 V, N = 40, h = 50 μm, g = 5 μm, the equation given in c) gives a force magnitude of 35.4 μN, giving a deflection either side of 35.4/18.8 = 1.89 μm and a rotation angle at the mirror of $\frac{3}{2} \times \frac{1.89}{500} = 0.0057$ or 0.32°. [4]